

Automated Laneway Home Feasibility Dashboard

Partner: Green Metrics Technology

Team: Jamie Breault, Sasivimol Sirijangkapattana, Yusen Rong, Zihao Sheng

PROBLEM OVERVIEW

Municipal zoning bylaws define critical rules for land use and development, including building height limits, setbacks, density, lot coverage, permitted uses, and accessory dwelling requirements. For laneway homes, these rules determine whether a structure is legally feasible on a specific parcel and what form it may take.

In practice, these rules are difficult to apply on a scale. Bylaws are published as long PDF documents, tables, amendments, and zoning schedules with different terminology and update cycles across municipalities. Checking whether a parcel can support a laneway home is still largely manual: a practitioner must identify the parcel, confirm its zoning, locate the relevant sections, interpret the rules, and translate them into spatial constraints.

The project focuses on a practical prototype: a dashboard that displays city lot information through a GIS map, allows a user to select a lot, and shows whether a laneway home can be built on that lot — with traceable evidence for relevant rules. Burnaby, Vancouver, and Calgary are prioritized as the first prototype cities, supporting future generalization.

MOTIVATION AND PURPOSE

The purpose of this project is to convert zoning bylaw information into a structured and verifiable rule layer that supports parcel-level feasibility analysis. It connects rule extraction, normalization, verification, GIS-based spatial analysis, and dashboard presentation.

- Reduce manual effort by allowing users to inspect lot-level laneway home feasibility directly on a map.
- Improve consistency by applying normalized rules rather than relying on repeated manual interpretation.
- Maintain transparency by storing the source evidence for each extracted rule and making it accessible in the dashboard.

WHAT WE BUILT

We built an end-to-end automated pipeline with four stages: Extract — a large language model reads the bylaw PDF and pulls out candidate rules (height, setbacks, lot coverage) from both text and complex tables; Verify — a separate checking system approves only rules with direct evidence from the source document, flagging uncertain cases for human review; GIS — verified rules are spatially applied to parcel data to define buildable areas and fit building footprints for each lot; Visualise — verified rules are combined with GIS parcel data and displayed in an interactive Felt-based dashboard where a user selects any lot and instantly sees the applicable rules and building constraints.

RESULTS

Once verified rules from the extraction pipeline are combined with GIS-calculated parcel attributes, the results are loaded into the dashboard. Any homeowner, developer, or planner can click on a lot and immediately see whether a laneway home is buildable — along with the applicable building constraints and the bylaw evidence behind each rule. No prior knowledge of zoning documents is required. The system approves only rules with sufficient evidence, ensuring everything shown in the dashboard can be trusted.

WHAT THIS MEANS

Zoning information that previously required hours of PDF searching is now a single click on a map. A homeowner sees what their lot allows. A planner scans an entire neighborhood. A developer filters parcels by buildable rules. The pipeline generalizes: adding a new city requires only a bylaw PDF and a configuration file, making it a scalable foundation for future expansion across BC municipalities.